



Abstract of Scientific Justification (300 words max) (2 points): Summarize your program. Why is this observing program scientifically interesting? What target(s) are you trying to observe and with what observational modes? What science will your program uncover? **We are observing two M33 cepheids to determine the distance to M33 via redshift scaling. This is scientifically interesting because it allows us to get hands-on experience with redshift, something that has only been a part of curriculum thus far. We're observing D33 J013401.2+303113.7 and D33 J013352.4+303844.2, two classical cepheid variables. We will observe them in the G-band and the I-band using the appropriate filters on the 90prime instrument in the Bok 90" telescope. The distance to M33 is well documented, so this will likely only confirm what is already known or help refine the value of its distance at the absolute most.**

Telescope Requested (0.5 points): **Bok 90", Kuiper cannot see in the G-band which we require**

Instrument (0.5 points): **We need the 90Prime instrument, specifically the g and i filters.**

Tolerable Moon Brightness (dark, grey, bright, etc) (0.5 points): Write out what moon brightness is acceptable for your program and why. How does the moon impact the sensitivity of your observations?

We will require dark skies for our program, the moon will drown out the cepheids in the g band. It may be possible to get the data in the i-band but it would take longer and would leave us with an incomplete data set to work with. The rayleigh scattering from moonlight won't be nearly as bad in the I-band due to it's sharp fall off, but it will still be irritating.

Target List (2 points)

List potential targets you may observe and their basic properties like brightness. Under visibility write what months/days the target is up. For example "Object can be observed Feb-March."

Object Name	Right Ascension	Declination	V magnitude	Object Type	Visibility
D33 J013401.2+303113.7	01 34 01.1961715224	+30 31 13.402671456	19.0 +/- 0.4	Classical Cepheid Variable	July(start) - Dec(start)
D33 J013352.4+303844.2	01 33 52.4402010666	+30 38 44.082599552	19.0 +/- 0.4	Classical Cepheid Variable	July(start) - Dec(start)

Experimental Design & Technical Description (4.5 points total)

How much time is required for your program and what signal-to-noise is necessary for your observations?

(2 points) SNR ~ 100 , at most we'd need 30 minutes for each filter. So assuming no down time between switching the filters we'd need approximately an hour on the telescope.

How will these observations contribute toward the accomplishment of the goals outlined in the abstract?

(2 points) These observations will allow us to add another data point to the M33 "sawtooth" magnitude plots seen in figure 2 of [this paper](#).

Back Up Plan. Describe a potential project that can be done with poor weather or archival data. **(0.5**

points) Plan B, stick to Plan A. Switch to I-band if the moon rises and we can't get the g-band data.